

NEVERMIND. Start with one moving link.

The first commit jumped straight to assigning beams in a frozen scene. That assumes I already know when a beam can serve a target. I'd rather work that out for one moving satellite first. One satellite, one ground target, one pass: orbit, where to point, when the signal arrives, and what Doppler does to its frequency. I also tried one nearby interfering beam to see what a toy SINR looks like.

[done-ish] Motion first. Keep it in a plane for now. Let $r = (x, y)$ and $v = (v_x, v_y)$ for the satellite:

$$\dot{r} = v, \quad \dot{v} = -\frac{\mu r}{\|r\|^3}.$$

Start on a circular orbit at $R + h$ with speed $\sqrt{\mu/(R + h)}$; take 1-second RK4 steps in C++. The code compares the final numerical position with the circular-orbit position at the same time. Ground target rotates:

$$g(t) = R(\cos \omega_E t, \sin \omega_E t).$$

All the calculation uses metres, seconds, radians. I picked $h = 550$ km and $f = 2$ GHz just to get numbers out.

[done] Aim + arrival. A signal sent from $r(t)$ has to meet the target at $g(t + \tau)$, not $g(t)$. I repeat this equation a few times to get τ :

$$\tau = \frac{\|g(t + \tau) - r(t)\|}{c}, \quad a = \frac{g(t + \tau) - r(t)}{\|g(t + \tau) - r(t)\|}.$$

a is the direction to aim; the code calculates it but does not draw a beam. It also measures how far the target moves if I aim at its old position $g(t)$; that output is `no_lead_m`. The `visible` column means elevation is at least 10° above the target's horizon. This is a point/ray toy model; I have not given the beam a width.

[done] Doppler. Satellite and target both move. In this model, with a pointing along the ray,

$$\frac{f_{\text{received}}}{f_{\text{sent}}} = \frac{c - a \cdot v_g(t + \tau)}{c - a \cdot v_s(t)}.$$

The output shows the shift if I send 2 GHz, plus the transmit adjustment needed to get 2 GHz at the target. Approaching gives a higher received frequency; receding gives a lower one.

Numbers so far: directly overhead: 550 km range, 1.835 ms travel, about 0.85 m of ground motion during that time. At $t = -120$ s the shift is about +37.5 kHz; at $t = +120$ s it is about -37.5 kHz. Small aiming miss, noticeable frequency shift. The size of that miss relative to a beam footprint comes later.

[added] **Beam contour (?) + SINR.** I put a second ground target 0.3° away and point another beam from the same satellite at it. Both beams use the same frequency and 10 W. I am pretending the transmit gain falls off like

$$G(\theta) = 1000 \exp \left[-\frac{1}{2} \left(\frac{\theta}{2^\circ} \right)^2 \right].$$

The receiver gain is 1. Same free-space path factor for both beams at the first target; noise is kTB with $T = 290$ K and $B = 1$ MHz:

$$S = 10G(0) \left(\frac{\lambda}{4\pi d} \right)^2, \quad I = 10G(\theta) \left(\frac{\lambda}{4\pi d} \right)^2, \quad \text{SINR} = \frac{S}{I + kTB}.$$

At overhead I get about 6.5 dB SINR. Near the edge of this pass it is around 0 dB. That is what this made-up contour says, not an antenna measurement. The code re-aims both beams at every sample, so there is no stale-pointing effect yet.

[next] **Updates???** Hold a pointing direction and transmit frequency for a few seconds, then see what goes bad before the next update. After that I can clean up the note and list the assumptions in one place.